SOLUTIONS: SYSTEMS IDENTIFICATION AND LEARNING

1. a) Firstly, using the delay operator z^{-1}, we get:

$$
\begin{aligned}
& y(t) = \frac{1}{4}w(t) + \frac{11}{30}w(t-1) + \frac{2}{15}w(t-2) - \frac{1}{4}y(t-1) + \frac{3}{8}y(t-2) \\
& \implies \left(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}\right)y(t) = \left(\frac{1}{4} + \frac{11}{30}z^{-1} + \frac{2}{15}z^{-2}\right)w(t) \\
& \implies y(t) = \frac{\frac{1}{4} + \frac{11}{30}z^{-1} + \frac{2}{15}z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}}w(t) = \underbrace{\frac{\frac{1}{4}z^2 + \frac{11}{30}z + \frac{2}{15}}{z^2 + \frac{1}{4}z - \frac{3}{8}}}_{G_1(z)}w(t)
\end{aligned}
$$

Analogously:

$$
\begin{aligned}
& w(t) = -\frac{9}{2}e(t) - 3e(t-1) + 4e(t-2) - \frac{13}{10}w(t-1) - \frac{2}{5}w(t-2) \\
& \implies \left(1 + \frac{13}{10}z^{-1} + \frac{2}{5}z^{-2}\right)w(t) = \left(-\frac{9}{2} - 3z^{-1} + 4z^{-2}\right)e(t) \\
& \implies w(t) = \frac{-\frac{9}{2} - 3z^{-1} + 4z^{-2}}{1 + \frac{13}{10}z^{-1} + \frac{2}{5}z^{-2}}e(t) = \underbrace{\frac{-\frac{9}{2}z^2 - 3z + 4}{z^2 + \frac{13}{10}z + \frac{2}{5}}}_{G_2(z)}e(t)
\end{aligned}
$$

The process $y(\cdot)$ is stationary in steady state if $w(\cdot)$ is stationary and if the poles of $G_1(\cdot)$ are located strictly inside the unit circle. Also, the process $w(\cdot)$ is stationary in steady state if $e(\cdot)$ is stationary and if the poles of $G_2(\cdot)$ located strictly inside the unit circle. Since $e(\cdot)$ is stationary by definition, $y(\cdot)$ is stationary in steady state if the poles of $G_1(\cdot)$ and $G_2(\cdot)$ are located strictly inside the unit circle.

The poles p_1 and p_2 of the transfer function $G_1(\cdot)$ are given by:

$$
G_1(\cdot): \quad \begin{aligned} p_1 &= +\frac{1}{2} \implies |p_1| = \frac{1}{2} < 1 \\ p_2 &= -\frac{3}{4} \implies |p_2| = \frac{3}{4} < 1 \end{aligned}
$$

The poles p_3 and p_4 of the transfer function $G_2(\cdot)$ are given by:

$$
G_2(\cdot): \quad \begin{aligned} p_3 &= -\frac{1}{2} \implies |p_3| = \frac{1}{2} < 1 \\ p_4 &= -\frac{4}{5} \implies |p_4| = \frac{4}{5} < 1 \end{aligned}
$$

Therefore, $y(\cdot)$ is a stationary process.

Since the process is stationary, we have:

$$
m_y = \mathbb{E}[y(t)] = G_1(1) \cdot G_2(1) \cdot \mathbb{E}[\eta(t)] = \frac{6}{7} \cdot \left(-\frac{35}{27}\right) \cdot \frac{2}{9} = -\frac{10}{9}
$$

[6 Marks]

Answers

b) Since the process $y(\cdot)$ is stationary, it is convenient to transform the transfer functions $G_1(\cdot)$ and $G_2(\cdot)$ determined in the answer to Question 1-a) into spectral canonical forms. We obtain:

$$\begin{aligned}
y(t) &= G_1(z)w(t) = G_1(z)G_2(z)e(t) \\
&= \frac{\frac{1}{4}z^2+\frac{11}{30}z+\frac{2}{15}}{z^2+\frac{1}{4}z-\frac{3}{8}}\,\frac{-\frac{9}{2}z^2-3z+4}{z^2+\frac{13}{10}z+\frac{2}{5}}e(t) \\
&= \frac{\frac{1}{4}\cancel{\left(z+\frac{4}{5}\right)}\left(z+\frac{2}{3}\right)\left(-\frac{9}{2}\right)\left(z+\frac{4}{3}\right)\left(z-\frac{2}{3}\right)}{\left(z-\frac{1}{2}\right)\left(z+\frac{3}{4}\right)\left(z+\frac{1}{2}\right)\cancel{\left(z+\frac{4}{5}\right)}}e(t) \\
&= -\frac{9}{2}\frac{1}{4}\frac{4}{3}\underbrace{\frac{3}{4}\frac{z+\frac{4}{3}}{z+\frac{3}{4}}}_{\text{All-pass filter}}\frac{\left(z+\frac{2}{3}\right)\left(z-\frac{2}{3}\right)}{\left(z-\frac{1}{2}\right)\left(z+\frac{1}{2}\right)}e(t) \\
&= -\frac{3}{2}\frac{z^2-\frac{4}{9}}{z^2-\frac{1}{4}}e(t) = \underbrace{\frac{1-\frac{4}{9}z^{-2}}{1-\frac{1}{4}z^{-2}}}_{H(z)}\eta(t) = H(z)\eta(t)
\end{aligned}$$

where we let

$$\eta(t) = -\frac{3}{2}e(t)$$

Therefore:

$$\eta(\cdot) \sim WN\left(-\frac{3}{2},\frac{1}{2}\right)$$

Accordingly, the process $y(\cdot)$ can be equivalently re-written in the much simpler form:

$$y(t) = \frac{1}{4}y(t-2)+\eta(t)-\frac{4}{9}\eta(t-2)$$

From the definition of auto-covariance function, we get

$$\gamma_y(\tau) = \mathbb{E}[(y(t)-m_y)(y(t-\tau)-m_y)] = \mathbb{E}[\tilde{y}(t)\tilde{y}(t-\tau)]$$

where

$$\begin{aligned}
\tilde{y}(t) &= y(t)-m_y = y(t)+\frac{10}{9} \\
&= \frac{1}{4}y(t-2)+\eta(t)-\frac{4}{9}\eta(t-2)-\left(-\frac{10}{9}\right) \\
&= \frac{1}{4}y(t-2)+\eta(t)-\frac{4}{9}\eta(t-2)-\left(\frac{1}{4}\left(-\frac{10}{9}\right)+\left(-\frac{3}{2}\right)-\frac{4}{9}\left(-\frac{3}{2}\right)\right) \\
&= \frac{1}{4}\left(y(t-2)+\frac{10}{9}\right)+\left(\eta(t)+\frac{3}{2}\right)-\frac{4}{9}\left(\eta(t-2)+\frac{3}{2}\right) \\
&= \frac{1}{4}\tilde{y}(t-2)+\tilde{\eta}(t)-\frac{4}{9}\tilde{\eta}(t-2)
\end{aligned}$$

where $\tilde{\eta}(t) = \eta(t)+\frac{3}{2}$. Therefore, $\tilde{\eta} \sim WN\left(0,\frac{1}{2}\right)$, and

$$\begin{aligned}
\gamma_y(\tau) &= \mathbb{E}[\tilde{y}(t)\tilde{y}(t-\tau)] \\
&= \mathbb{E}\left[\left(\frac{1}{4}\tilde{y}(t-2)+\tilde{\eta}(t)-\frac{4}{9}\tilde{\eta}(t-2)\right)\tilde{y}(t-\tau)\right] \\
&= \frac{1}{4}\mathbb{E}[\tilde{y}(t-2)\tilde{y}(t-\tau)]+\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t-\tau)]-\frac{4}{9}\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t-\tau)]
\end{aligned}$$

Answers

In the case with $\tau = 0$, $\tau = 1$ and $\tau = 2$, we have

$$\gamma_y(0) = \frac{1}{4}\underbrace{\mathbb{E}[\tilde{y}(t-2)\tilde{y}(t)]}_{\gamma_y(2)} + \mathbb{E}[\tilde{\eta}(t)\tilde{y}(t)] - \frac{4}{9}\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t)]$$

$$\gamma_y(1) = \frac{1}{4}\underbrace{\mathbb{E}[\tilde{y}(t-2)\tilde{y}(t-1)]}_{\gamma_y(1)} + \underbrace{\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t-1)]}_{0} - \frac{4}{9}\underbrace{\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t-1)]}_{\mathbb{E}[\tilde{\eta}(t-1)\tilde{y}(t)]}$$

$$\gamma_y(2) = \frac{1}{4}\underbrace{\mathbb{E}[\tilde{y}(t-2)\tilde{y}(t-2)]}_{\gamma_y(0)} + \underbrace{\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t-2)]}_{0} - \frac{4}{9}\underbrace{\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t-2)]}_{\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t)]}$$

where

$$\begin{aligned}\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t)] &= \mathbb{E}\left[\tilde{\eta}(t)\left(\frac{1}{4}\tilde{y}(t-2) + \tilde{\eta}(t) - \frac{4}{9}\tilde{\eta}(t-2)\right)\right] \\ &= \frac{1}{4}\underbrace{\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t-2)]}_{0} + \underbrace{\mathbb{E}\left[\tilde{\eta}(t)^2\right]}_{\frac{1}{2}} - \frac{4}{9}\underbrace{\mathbb{E}[\tilde{\eta}(t)\tilde{\eta}(t-2)]}_{0} = \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\tilde{\eta}(t-1)\tilde{y}(t)] &= \mathbb{E}\left[\tilde{\eta}(t-1)\left(\frac{1}{4}\tilde{y}(t-2) + \tilde{\eta}(t) - \frac{4}{9}\tilde{\eta}(t-2)\right)\right] \\ &= \frac{1}{4}\underbrace{\mathbb{E}[\tilde{\eta}(t-1)\tilde{y}(t-2)]}_{0} + \underbrace{\mathbb{E}[\tilde{\eta}(t-1)\tilde{\eta}(t)]}_{0} - \frac{4}{9}\underbrace{\mathbb{E}[\tilde{\eta}(t-1)\tilde{\eta}(t-2)]}_{0} = 0\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t)] &= \mathbb{E}\left[\tilde{\eta}(t-2)\left(\frac{1}{4}\tilde{y}(t-2) + \tilde{\eta}(t) - \frac{4}{9}\tilde{\eta}(t-2)\right)\right] \\ &= \frac{1}{4}\underbrace{\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t-2)]}_{\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t)]=\frac{1}{2}} + \underbrace{\mathbb{E}[\tilde{\eta}(t-2)\tilde{\eta}(t)]}_{0} - \frac{4}{9}\underbrace{\mathbb{E}[\tilde{\eta}(t-2)\tilde{\eta}(t-2)]}_{\frac{1}{2}} = -\frac{7}{72}\end{aligned}$$

Hence:

$$\gamma_y(0) = \gamma_y(2) + \frac{1}{2} + \frac{7}{162}; \quad \gamma_y(1) = \frac{1}{4}\gamma_y(1); \quad \gamma_y(2) = \frac{1}{4}\gamma_y(0) - \frac{2}{9}$$

From the second equation, we notice that $\gamma_y(1) = 0$. Then, substituting the third equation in the first one, we have

$$\gamma_y(0) = \frac{1}{4}\gamma_y(0) - \frac{2}{9} + \frac{1}{2} + \frac{7}{162} \implies \gamma_y(0) = \frac{-\frac{2}{9} + \frac{1}{2} + \frac{7}{162}}{1 - \frac{1}{4}} = \frac{104}{243} \approx 0.43$$

From the third equation, we also obtain

$$\gamma_y(2) = \frac{1}{4}\frac{104}{243} - \frac{2}{9} = -\frac{28}{243} \approx 0.12$$

Then, given a generic $\tau > 2$, we have

$$\gamma_y(\tau) = \frac{1}{4}\underbrace{\mathbb{E}[\tilde{y}(t-2)\tilde{y}(t-\tau)]}_{\gamma_y(\tau-2)} + \underbrace{\mathbb{E}[\tilde{\eta}(t)\tilde{y}(t-\tau)]}_{0} - \frac{4}{9}\underbrace{\mathbb{E}[\tilde{\eta}(t-2)\tilde{y}(t-\tau)]}_{0} = \frac{1}{4}\gamma_y(\tau-2)$$

Since the auto-covariance function is even, we conclude by stating that

$$\gamma_y(\tau) = \begin{cases} 104/243 & \text{If } \tau = 0 \\ 0 & \text{If } |\tau| = 1 \\ -28/243 & \text{If } |\tau| = 2 \\ \frac{1}{4}\gamma_y(|\tau| - 2) & \text{If } |\tau| > 2 \end{cases}$$

[7 Marks]

Answers

c) From the answer to Question 1-b) we have:

$$y(t) = H(z)\eta(t) = \frac{1-\frac{4}{9}z^{-2}}{1-\frac{1}{4}z^{-2}} \cdot \eta(t)$$

where

$$\eta(\cdot) \sim WN\left(-\frac{3}{2}, \frac{1}{2}\right)$$

Therefore:

$$\begin{aligned}
\Gamma_y(\omega) &= \left|H\left(e^{j\omega}\right)\right|^2 \cdot \Gamma_\eta(\omega) \\
&= \left|\frac{1-\frac{4}{9}e^{-2j\omega}}{1-\frac{1}{4}e^{-2j\omega}}\right|^2 \cdot \frac{1}{2} \\
&= \frac{\left(1-\frac{4}{9}e^{-2j\omega}\right)\left(1-\frac{4}{9}e^{+2j\omega}\right)}{\left(1-\frac{1}{4}e^{-2j\omega}\right)\left(1-\frac{1}{4}e^{+2j\omega}\right)} \cdot \frac{1}{2} \\
&= \frac{1-\frac{4}{9}e^{-2j\omega}-\frac{4}{9}e^{+2j\omega}+\frac{4^2}{9^2}}{1-\frac{1}{4}e^{-2j\omega}-\frac{1}{4}e^{+2j\omega}+\frac{1}{4^2}} \cdot \frac{1}{2} \\
&= \frac{\frac{97}{81}-\frac{4}{9}\left(e^{-2j\omega}+e^{+2j\omega}\right)}{\frac{17}{16}-\frac{1}{4}(e^{-2j\omega}+e^{+2j\omega})} \cdot \frac{1}{2} \\
&= \frac{\frac{97}{81}-\frac{8}{9}\cos(2\omega)}{\frac{17}{16}-\frac{1}{2}\cos(2\omega)} \cdot \frac{1}{2} \\
&= \frac{1552-1152\cos(2\omega)}{2754-1296\cos(2\omega)}
\end{aligned}$$

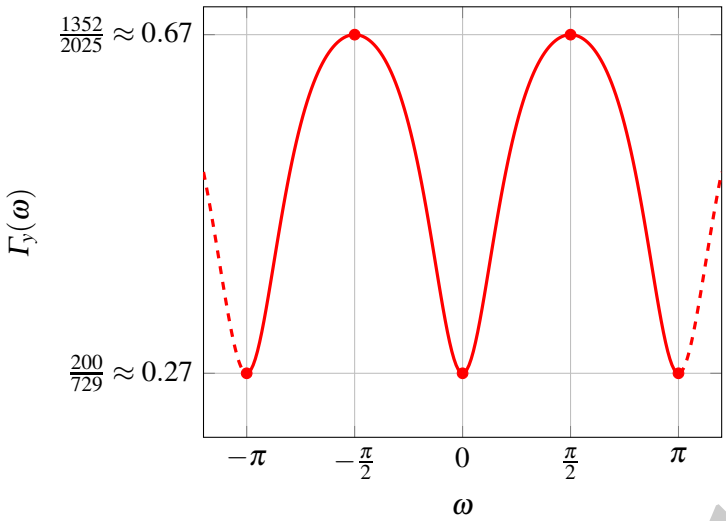
[4 Marks]

d) To sketch the diagram of $\Gamma_y(\omega)$ we compute some value of $\Gamma_y(\cdot)$ to interpolate them. To select the points, it is useful to remember that $\Gamma_y(\cdot)$ is a periodic function with period 2π and that is an even function. Thus:

$$\begin{aligned}
\Gamma_y(0) &= \frac{1552-1152\cos(0)}{2754-1296\cos(0)} = \frac{200}{729} \approx 0.27 \\
\Gamma_y\left(\frac{\pi}{2}\right) = \Gamma_y\left(-\frac{\pi}{2}\right) &= \frac{1552-1152\cos(\pi)}{2754-1296\cos(\pi)} = \frac{1352}{2025} \approx 0.67 \\
\Gamma_y(\pi) = \Gamma_y(-\pi) &= \frac{1552-1152\cos(2\pi)}{2754-1296\cos(2\pi)} = \frac{200}{729} \approx 0.27
\end{aligned}$$

The diagram of $\Gamma_y(\omega)$ is given in figure below.

Answers



[3 Marks]

Answers

2. a) First, we show that the process $y(\cdot)$ is stationary in steady state. According to the recursive equation (2.1) in the text of the exam paper, we obtain:

$$\begin{aligned} y(t) &= \frac{1}{2}y(t-1) - \frac{1}{8}y(t-2) + w(t-2) + \frac{2}{3}w(t-3) \\ \Longrightarrow y(t)&\left(1 - \frac{1}{2}z^{-1} + \frac{1}{8}z^{-2}\right) = w(t)\left(z^{-2} + \frac{2}{3}z^{-3}\right) \\ \Longrightarrow y(t) &= \frac{z^{-2} + \frac{2}{3}z^{-3}}{1 - \frac{1}{2}z^{-1} + \frac{1}{8}z^{-2}}w(t) = \frac{z + \frac{2}{3}}{z^3 - \frac{1}{2}z^2 + \frac{1}{8}z}w(t) \end{aligned}$$

Due to linearity, the stochastic process $y(\cdot)$ is stationary if $w(\cdot)$ is also stationary and the dynamical system

$$y(t) = \frac{z + \frac{2}{3}}{z^3 - \frac{1}{2}z^2 + \frac{1}{8}z}w(t)$$

is asymptotically stable. The process $w(\cdot)$ is stationary by the definition of white process. Furthermore, the above dynamical system has three poles:

$$p_1 = 0; \quad p_2 = \frac{1}{4} + j\frac{1}{4}; \quad p_3 = \frac{1}{4} - j\frac{1}{4}$$

Since

$$|p_2| = |p_3| = \frac{\sqrt{2}}{4} \approx 0.35 < 1$$

then the above dynamical system is asymptotically stable and thus the process $y(\cdot)$ is stationary.

Now, we notice that the process $y(\cdot)$ is 0-mean because it is the output of a linear dynamical system fed with a 0-mean process. Hence, the variance of the process is given by

$$\begin{aligned} \gamma_y(0) = \mathbb{E}\left[y(t)^2\right] &= \mathbb{E}\left[\left(\frac{1}{2}y(t-1) - \frac{1}{8}y(t-2) + w(t-2) + \frac{2}{3}w(t-3)\right)^2\right] \\ &= \frac{1}{4}\underbrace{\mathbb{E}\left[y(t-1)^2\right]}_{\gamma_y(0)} + \frac{1}{64}\underbrace{\mathbb{E}\left[y(t-2)^2\right]}_{\gamma_y(0)} + \underbrace{\mathbb{E}\left[w(t-2)^2\right]}_{1} + \frac{4}{9}\underbrace{\mathbb{E}\left[w(t-3)^2\right]}_{1} \\ &\quad - \frac{1}{8}\underbrace{\mathbb{E}[y(t-1)y(t-2)]}_{\gamma_y(1)} + \underbrace{\mathbb{E}[y(t-1)w(t-2)]}_{\mathbb{E}[y(t)w(t-1)]} + \frac{2}{3}\underbrace{\mathbb{E}[y(t-1)w(t-3)]}_{\mathbb{E}[y(t)w(t-2)]} \\ &\quad - \frac{1}{4}\underbrace{\mathbb{E}[y(t-2)w(t-2)]}_{\mathbb{E}[y(t)w(t)]} - \frac{1}{6}\underbrace{\mathbb{E}[y(t-2)w(t-3)]}_{\mathbb{E}[y(t)w(t-1)]} + \frac{4}{3}\underbrace{\mathbb{E}[w(t-2)w(t-3)]}_{0} \\ &= \frac{1}{4}\gamma_y(0) + \frac{1}{64}\gamma_y(0) + 1 + \frac{4}{9} - \frac{1}{8}\gamma_y(1) + \mathbb{E}[y(t)w(t-1)] + \frac{2}{3}\mathbb{E}[y(t)w(t-2)] \\ &\quad - \frac{1}{4}\mathbb{E}[y(t)w(t)] - \frac{1}{6}\mathbb{E}[y(t)w(t-1)] \end{aligned}$$

where the various terms in the above summation can be computed as follows:

$$\begin{aligned} \mathbb{E}[y(t)w(t)] &= \mathbb{E}\left[\left(\frac{1}{2}y(t-1) - \frac{1}{8}y(t-2) + w(t-2) + \frac{2}{3}w(t-3)\right)w(t)\right] \\ &= \frac{1}{2}\underbrace{\mathbb{E}[y(t-1)w(t)]}_{0} - \frac{1}{8}\underbrace{\mathbb{E}[y(t-2)w(t)]}_{0} + \underbrace{\mathbb{E}[w(t-2)w(t)]}_{0} \end{aligned}$$

Answers

$$+\frac{2}{3}\underbrace{\mathbb{E}[w(t-3)w(t)]}_{0}=0$$

$$\begin{aligned}\mathbb{E}[y(t)w(t-1)] &= \mathbb{E}\left[\left(\frac{1}{2}y(t-1)-\frac{1}{8}y(t-2)+w(t-2)+\frac{2}{3}w(t-3)\right)w(t-1)\right]\\ &= \frac{1}{2}\underbrace{\mathbb{E}[y(t-1)w(t-1)]}_{\mathbb{E}[y(t)w(t)]=0}-\frac{1}{8}\underbrace{\mathbb{E}[y(t-2)w(t-1)]}_{0}+\underbrace{\mathbb{E}[w(t-2)w(t-1)]}_{0}\\ &\quad+\frac{2}{3}\underbrace{\mathbb{E}[w(t-3)w(t-1)]}_{0}=0\end{aligned}$$

$$\begin{aligned}\mathbb{E}[y(t)w(t-2)] &= \mathbb{E}\left[\left(\frac{1}{2}y(t-1)-\frac{1}{8}y(t-2)+w(t-2)+\frac{2}{3}w(t-3)\right)w(t-2)\right]\\ &= \frac{1}{2}\underbrace{\mathbb{E}[y(t-1)w(t-2)]}_{\mathbb{E}[y(t)w(t-1)]=0}-\frac{1}{8}\underbrace{\mathbb{E}[y(t-2)w(t-2)]}_{\mathbb{E}[y(t)w(t)]=0}+\underbrace{\mathbb{E}\left[w(t-2)^2\right]}_{1}\\ &\quad+\frac{2}{3}\underbrace{\mathbb{E}[w(t-3)w(t-2)]}_{0}=1\end{aligned}$$

Moreover, we have

$$\begin{aligned}\gamma_y(1)=\mathbb{E}[y(t)y(t-1)] &= \mathbb{E}\left[\left(\frac{1}{2}y(t-1)-\frac{1}{8}y(t-2)+w(t-2)+\frac{2}{3}w(t-3)\right)y(t-1)\right]\\ &= \frac{1}{2}\underbrace{\mathbb{E}\left[y(t-1)^2\right]}_{\gamma_y(0)}-\frac{1}{8}\underbrace{\mathbb{E}[y(t-2)y(t-1)]}_{\gamma_y(1)}+\underbrace{\mathbb{E}[w(t-2)y(t-1)]}_{\mathbb{E}[y(t)w(t-1)]=0}+\frac{2}{3}\underbrace{\mathbb{E}[w(t-3)y(t-1)]}_{\mathbb{E}[y(t)w(t-2)]=1}\\ &= \frac{1}{2}\gamma_y(0)-\frac{1}{8}\gamma_y(1)+\frac{2}{3}\end{aligned}$$

Therefore, we need to solve the linear system of equations

$$\begin{cases}\gamma_y(0)=\frac{1}{4}\gamma_y(0)+\frac{1}{64}\gamma_y(0)+1+\frac{4}{9}-\frac{1}{8}\gamma_y(1)+\frac{2}{3}\\ \gamma_y(1)=\frac{1}{2}\gamma_y(0)-\frac{1}{8}\gamma_y(1)+\frac{2}{3}\end{cases}$$

That has solution

$$\gamma_y(0)=\frac{704}{273}\approx 2.58 \qquad\qquad \gamma_y(1)=\frac{1424}{819}\approx 1.74$$

[8 Marks]

b) We notice that $\mathscr{M}_1$ is a family of AR(1) models. Hence, the one-step optimal predictor is given by

$$\hat{y}(t|t-1)=ay(t-1)$$

Since we have an arbitrary large number of data, according to the PEM criterion, the optimal estimated value of a is obtained by minimizing the cost function

$$J(a)=\mathbb{E}\left[\left(y(t)-\hat{y}(t|t-1)\right)^2\right]$$

Answers

$$
\begin{aligned}
&= \mathbb{E}\Big[\big(y(t) - ay(t-1)\big)^2\Big] \\
&= \mathbb{E}\big[y(t)^2\big] - 2a\mathbb{E}[y(t)y(t-1)] + a^2\mathbb{E}\big[y(t-1)^2\big] \\
&= \gamma_y(0) - 2a\gamma_y(1) + a^2\gamma_y(0) \\
&= \frac{704}{273} - a\frac{2848}{819} + a^2\frac{704}{273}
\end{aligned}
$$

To find the minimum of J, we compute its stationary point as follows:

$$
\frac{d}{da}J(a) = -\frac{2848}{819} + a\frac{1408}{273} = 0 \implies a^\circ = \frac{\frac{2848}{819}}{\frac{1408}{273}} = \frac{89}{132} \approx 0.67
$$

To find the optimal value of λ^2, we evaluate the cost function in a°. Therefore, we have

$$
\begin{aligned}
\lambda^{2^\circ} &= J(a^\circ) \\
&= \frac{704}{273} - \frac{89}{132}\cdot\frac{2848}{819} + \left(\frac{89}{132}\right)^2\cdot\frac{704}{273} = \frac{2924}{2079} \approx 1.41
\end{aligned}
$$

[6 Marks]

c) We notice that $\mathscr{M}_2$ is a family of AR(2) models. Hence, the one-step optimal predictor is given by

$$
\hat{y}(t|t-1) = by(t-2)
$$

Since we have an arbitrary large number of data, according to the PEM criterion, the optimal estimated value of b is obtained by minimizing the cost function

$$
\begin{aligned}
J(b) &= \mathbb{E}\Big[\big(y(t) - \hat{y}(t|t-2)\big)^2\Big] \\
&= \mathbb{E}\Big[\big(y(t) - by(t-2)\big)^2\Big] \\
&= \mathbb{E}\big[y(t)^2\big] - 2b\mathbb{E}[y(t)y(t-2)] + b^2\mathbb{E}\big[y(t-2)^2\big] \\
&= \gamma_y(0) - 2b\gamma_y(2) + b^2\gamma_y(0)
\end{aligned}
$$

where

$$
\begin{aligned}
\gamma_y(2) = \mathbb{E}[y(t)y(t-2)] &= \mathbb{E}\left[\left(\frac{1}{2}y(t-1) - \frac{1}{8}y(t-2) + w(t-2) + \frac{2}{3}w(t-3)\right)y(t-2)\right] \\
&= \frac{1}{2}\underbrace{\mathbb{E}[y(t-1)y(t-2)]}_{\gamma_y(1)=\frac{1424}{819}} - \frac{1}{8}\underbrace{\mathbb{E}[y(t-2)y(t-2)]}_{\gamma_y(0)=\frac{704}{273}} + \underbrace{\mathbb{E}[w(t-2)y(t-2)]}_{\mathbb{E}[y(t)w(t)]=0} + \frac{2}{3}\underbrace{\mathbb{E}[w(t-3)y(t-2)]}_{\mathbb{E}[y(t)w(t-1)]=0} \\
&= \frac{1}{2}\cdot\frac{704}{273} - \frac{1}{8}\frac{704}{273} = \frac{64}{117} \approx 0.55
\end{aligned}
$$

Hence, we have

$$
J(b) = \frac{704}{273} - b\frac{128}{117} + b^2\frac{704}{273}
$$

To find the minimum of J, we compute its stationary point as follows:

$$
\frac{d}{db}J(b) = -\frac{128}{117} + b\frac{1408}{273} = 0 \implies b^\circ = \frac{\frac{128}{117}}{\frac{1408}{273}} = \frac{7}{33} \approx 0.21
$$

Answers

To find the optimal value of μ^2, we evaluate the cost function in b°. Therefore, we have

$$\begin{aligned}\mu^{2^\circ} &= J(b^\circ) \\ &= \frac{704}{273} - \frac{7}{33}\cdot\frac{128}{117} + \left(\frac{7}{33}\right)^2\frac{704}{273} = \frac{5120}{2079} \approx 2.46\end{aligned}$$

[6 Marks]

Answers

3. a) Working in terms of transfer functions in the $\mathscr{Z}$ transforms domain via an appropriate usage of the delay operator z^{-1}, the two recursive equations

$$\alpha(t+1) = \frac{3}{4}\alpha(t) + \frac{5}{6}\beta(t) + \frac{3}{4}e(t) + 2u(t)$$
$$\beta(t+1) = \frac{1}{3}\beta(t) + \frac{3}{2}e(t) + 3u(t)$$

can be rewritten as

$$\alpha(t+1) = \frac{3}{4}z^{-1}\alpha(t+1) + \frac{5}{6}\beta(t) + \frac{3}{4}e(t) + 2u(t)$$
$$\beta(t+1) = \frac{1}{3}z^{-1}\beta(t+1) + \frac{3}{2}e(t) + 3u(t)$$

or, equivalently

$$z\alpha(t) = \frac{3}{4}\alpha(t) + \frac{5}{6}\beta(t) + \frac{3}{4}e(t) + 2u(t)$$
$$z\beta(t) = \frac{1}{3}\beta(t) + \frac{3}{2}e(t) + 3u(t)$$

Hence, simple algebra gives:

$$\alpha(t) = \underbrace{\frac{\frac{3}{4}z+1}{(z-3/4)(z-1/3)}e(t)}_{\alpha_{\mathrm{e}}(t)} + \underbrace{\frac{2z+\frac{11}{6}}{(z-3/4)(z-1/3)}u(t)}_{\alpha_{\mathrm{u}}(t)}$$
$$\beta(t) = \underbrace{\frac{3/2}{z-1/3}e(t)}_{\beta_{\mathrm{e}}(t)} + \underbrace{\frac{3}{z-1/3}u(t)}_{\beta_{\mathrm{u}}(t)}$$

Both processes are the sum of two independent processes. Therefore, $\alpha(\cdot)$ is stationary if $\alpha_{\mathrm{e}}(\cdot)$ and $\alpha_{\mathrm{u}}(t)$ are stationary. Similarly, $\beta(\cdot)$ is stationary if $\beta_{\mathrm{e}}(\cdot)$ and $\beta_{\mathrm{u}}(t)$ are stationary.

Since $e(\cdot)$ is a stationary process, $\alpha_{\mathrm{e}}(\cdot)$ and $\beta_{\mathrm{e}}(\cdot)$ are stationary if the corresponding transfer function have poles strictly inside the unit circle. Since the poles are

$$p_1 = \frac{3}{4}; \quad p_2 = \frac{1}{3}$$

we conclude that $\alpha_{\mathrm{e}}(\cdot)$ and $\beta_{\mathrm{e}}(\cdot)$ are stationary.

Analogously, $\alpha_{\mathrm{u}}(\cdot)$ and $\beta_{\mathrm{u}}(\cdot)$ are stationary processes if $u(\cdot)$ is stationary.

In particular, we have:

Case 1: Since $u_1(\cdot)$ is a white process, $\alpha_{\mathrm{u}}(\cdot)$ and $\beta_{\mathrm{u}}(\cdot)$ are stationary with $u(\cdot) = u_1(\cdot)$.

Case 2: At steady-state, $u_2(\cdot) = 0$ therefore $\alpha_{\mathrm{u}}(t) = \beta_{\mathrm{u}}(t) = 0$ and they are stationary.

Case 3: Since $u_3(\cdot)$ is a deterministic time-varying input (a sine wave) that does not vanish at steady-state, the expected value of the process is not constant over time, and thus the process is not stationary.

To summarize, $\alpha(\cdot)$ and $\beta(\cdot)$ are stationary in Case 1 and Case 2. Instead, they are not stationary in case 3.

[5 Marks]

Answers

b) With $u(t) = 0$, we have

$$\alpha(t) = \alpha_e(t) = \frac{\frac{3}{4}z+1}{(z-3/4)(z-1/3)}e(t) = \frac{18z+24}{24z^2-26z+6}e(t)$$

Firstly, we can note that the numerator has a root outside the unit circle. In particular, with $z = -\frac{4}{3}$ the numerator is equal to 0. For this reason, we apply an all-pass filter:

$$\begin{aligned}\alpha_e(t) &= \underbrace{\frac{4}{3}\frac{z+\frac{3}{4}}{z+\frac{4}{3}}}_{\text{All-pass filter}} \frac{18z+24}{24z^2-26z+6}e(t) \\ &= \frac{4}{3}\frac{z+\frac{3}{4}}{\cancel{z+\frac{4}{3}}}\frac{18\cancel{\left(z+\frac{4}{3}\right)}}{24z^2-26z+6}e(t) \\ &= \frac{12z+9}{12z^2-13z+3}e(t)\end{aligned}$$

Now, we collect the highest degree term in both the numerator and the denominator to make them monic, thus obtaining

$$\alpha_e(t) = \frac{1+\frac{3}{4}z^{-1}}{1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}}\frac{12z}{12z^2}e(t).$$

Finally, we introduce the new process

$$\eta(t) = \frac{12z}{12z^2}e(t) = e(t-1) \implies \eta \sim WN\left(0, \frac{1}{4}\right).$$

Hence, the spectral canonical form of $\alpha(\cdot)$ is given by:

$$\alpha_e(t) = \frac{1+\frac{3}{4}z^{-1}}{1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}}\eta(t)$$

and the corresponding recursive equation reads

$$\begin{aligned}\alpha_e(t)\left(1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}\right) &= \eta(t)\left(1+\frac{3}{4}z^{-1}\right) \\ \alpha_e(t) &= \frac{13}{12}\alpha_e(t-1) - \frac{1}{4}\alpha_e(t-2) + \eta(t) + \frac{3}{4}\eta(t-1).\end{aligned}$$

[4 Marks]

c) To determine the two steps ahead predictor, we need to use the spectral canonical form obtained in the answer to Question 3-b), that is:

$$\alpha_e(t) = \frac{1+\frac{3}{4}z^{-1}}{1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}}\eta(t)$$

The process $\alpha(\cdot)$ is an ARMA(2,1) process with polynomials $A(z)$ and $C(z)$ given by

$$A(z) = 1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}, \qquad C(z) = 1+\frac{3}{4}z^{-1}.$$

Now, it is possible to proceed with the long division to separate the "predictable" and the"unpredictable" parts:

Answers

$$
\begin{array}{rlll|l}
C(z) = & 1 & +\frac{3}{4}z^{-1} & & 1 \quad -\frac{13}{12}z^{-1} \quad +\frac{1}{4}z^{-2} \quad = A(z) \\
\hline
 & 1 & +\frac{13}{12}z^{-1} & -\frac{1}{4}z^{-2} & 1 \quad +\frac{11}{6}z^{-1} \quad = Q_2(z) \\
\hline
 & / & \frac{11}{6}z^{-1} & -\frac{1}{4}z^{-2} & \\
 & & -\frac{11}{6}z^{-1} & \frac{143}{72}z^{-2} \quad -\frac{11}{24}z^{-3} & \\
\hline
R_2(z) = & & / & \frac{125}{72}z^{-2} \quad -\frac{11}{24}z^{-3} &
\end{array}
$$

Then, the optimal predictor is given by

$$\widehat{\alpha}_{\mathrm{e}}(t|t-2) = \frac{R_2(z)}{C(z)}\alpha(t) = \frac{\frac{125}{72}z^{-2} - \frac{11}{24}z^{-3}}{1+\frac{3}{4}z^{-1}}\alpha(t) = \frac{\frac{125}{72} - \frac{11}{24}z^{-1}}{1+\frac{3}{4}z^{-1}}\alpha(t-2)$$

and the corresponding difference equation reads

$$\widehat{\alpha}_{\mathrm{e}}(t|t-2) = \frac{125}{72}\alpha_{\mathrm{e}}(t-2) - \frac{11}{24}\alpha_{\mathrm{e}}(t-3) - \frac{3}{4}\widehat{\alpha}_{\mathrm{e}}(t-1|t-3).$$

[4 Marks]

d) Now, $u(\cdot)$ is a deterministic function of time described as:

$$\alpha(t) = \frac{1+\frac{3}{4}z^{-1}}{1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}}\eta(t) + \frac{2z+\frac{11}{6}}{(z-3/4)(z-1/3)}u(t)$$

where the stochastic part of the process is written in terms of the spectral canonical form obtained in the answer to Question 3-b).

Then, we collect the highest degree term from the denominator of the second transfer function to obtain the more standard form

$$\alpha(t) = \frac{1+\frac{3}{4}z^{-1}}{1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}}\eta(t) + \frac{2z^{-1}+\frac{11}{6}z^{-2}}{1-\frac{13}{12}z^{-1}+\frac{1}{4}z^{-2}}u(t).$$

Therefore, $\alpha(\cdot)$ is ARMAX process with polynomials

The process $\alpha(\cdot)$ is an ARMAX(2,2,1) process with polynomials $A(z)$, $B(z)$, and $C(z)$ given by

$$A(z) = 1 - \frac{13}{12}z^{-1} + \frac{1}{4}z^{-2}, \quad B(z) = 2z^{-1} + \frac{11}{6}z^{-2}, \quad C(z) = 1 + \frac{3}{4}z^{-1}.$$

Since the stochastic part is the same of the process analyzed in the previous point, we already know the result of the long division:

$$R_2(z) = \frac{125}{72}z^{-2} - \frac{11}{24}z^{-3}, \qquad Q_2(z) = 1 + \frac{11}{6}z^{-1}.$$

Then, the optimal predictor is

$$\begin{aligned}
\widehat{\alpha}(t|t-2) &= \frac{R_2(z)}{C(z)}y(t) + \frac{B(z)Q_2(z)}{C(z)}u(t) \\
&= \frac{\frac{125}{72}z^{-2} - \frac{11}{24}z^{-3}}{1+\frac{3}{4}z^{-1}}\alpha(t) + \frac{\left(\frac{125}{72}z^{-2} - \frac{11}{24}z^{-3}\right)\left(1+\frac{11}{6}z^{-1}\right)}{1+\frac{3}{4}z^{-1}}u(t)
\end{aligned}$$

Answers

$$= \frac{\frac{125}{72} - \frac{11}{24}z^{-1}}{1+\frac{3}{4}z^{-1}}\alpha(t-2) + \frac{2z^{-1} + \frac{11}{2}z^{-2} + \frac{121}{36}z^{-3}}{1+\frac{3}{4}z^{-1}}u(t)$$

Therefore, the difference equation of the optimal two-steps predictor is

$$\widehat{\alpha}(t|t-2) = \frac{125}{72}\alpha(t-2) - \frac{11}{24}\alpha(t-3) + 2u(t-1) + \frac{11}{2}u(t-2) + \frac{121}{36}u(t-3) - \frac{3}{4}\widehat{\alpha}(t-1|t-3).$$

[4 Marks]

e) Taking into consideration the answers to Question 3-c) and Question 3-d), it turns out that

$$\varepsilon(t) = \alpha_{\mathrm{e}}(t) - \hat{\alpha}_{\mathrm{e}}(t|t-2) = \alpha(t) - \hat{\alpha}(t|t-2) = Q_2(z)\eta(t).$$

Therefore, we have

$$\varepsilon(t) = \left(1 + \frac{11}{6}z^{-1}\right)\eta(t) = \eta(t) + \frac{11}{6}\eta(t-1),$$

which is an MA process with variance

$$\mathbb{E}\left[\varepsilon(t)^2\right] = \mathbb{E}\left[\eta(t)^2\right]\left(1 + \left(\frac{11}{6}\right)^2\right) = \frac{1}{4}\left(1 + \left(\frac{11}{6}\right)^2\right) = \frac{157}{144} \approx 1.09$$

The variances of the two predictors are the same because the uncertainty of the prediction only depend on the stochastic part of the process. The only difference between the processes $\alpha_{\mathrm{e}}(\cdot)$ in Question 3-c), and $\alpha(\cdot)$, used in Question 3-d), is the effect of a deterministic input that does not change the accuracy of the prediction.

[3 Marks]

Answers

4. a) The estimation error is given by

$$\begin{aligned}\varepsilon &= \widehat{T} - T \\ &= (\alpha a + \beta b + \gamma) - (x + y) \\ &= \alpha\left(\frac{1}{3}x + \frac{2}{3}y + e_a\right) + \beta\left(\frac{4}{3}x + \frac{1}{2}y + e_b\right) + \gamma - x - y \\ &= x\left(\frac{\alpha}{3} + \frac{4\beta}{3} - 1\right) + y\left(\frac{2\alpha}{3} + \frac{\beta}{2} - 1\right) + \gamma + \alpha e_a + \beta e_b\end{aligned}$$

Therefore, the expected value of the estimation error reads

$$\begin{aligned}\mathbb{E}[\varepsilon] &= \mathbb{E}\left[x\left(\frac{\alpha}{3} + \frac{4\beta}{3} - 1\right) + y\left(\frac{2\alpha}{3} + \frac{\beta}{2} - 1\right) + \gamma + \alpha e_a + \beta e_b\right] \\ &= \mathbb{E}[x]\left(\frac{\alpha}{3} + \frac{4\beta}{3} - 1\right) + \mathbb{E}[y]\left(\frac{2\alpha}{3} + \frac{\beta}{2} - 1\right) + \gamma + \alpha \underbrace{\mathbb{E}[e_a]}_{0} + \beta \underbrace{\mathbb{E}[e_b]}_{1} \\ &= \mathbb{E}[x]\left(\frac{\alpha}{3} + \frac{4\beta}{3} - 1\right) + \mathbb{E}[y]\left(\frac{2\alpha}{3} + \frac{\beta}{2} - 1\right) + \gamma + \beta\end{aligned}$$

Thus, the estimator is unbiased for every x and y if the following conditions are satisfied:

$$\begin{cases} \dfrac{\alpha}{3} + \dfrac{4\beta}{3} - 1 = 0 \\ \dfrac{2\alpha}{3} + \dfrac{\beta}{2} - 1 = 0 \\ \gamma + \beta = 0 \end{cases}$$

Simple algebra yields the unique solution:

$$\alpha = \frac{15}{13}, \qquad \beta = \frac{6}{13}, \qquad \gamma = -\frac{6}{13}.$$

Hence, the unique unbiased estimator is given by

$$\widehat{T} = \frac{15}{13}a + \frac{6}{13}b - \frac{6}{13} = \frac{15}{13}a + \frac{6}{13}(b-1).$$

[7 Marks]

b) Owing to the answer to Question 4-a), there is only one unbiased estimator of the form defined in Question 4-a). Hence, necessarily, the estimator

$$\widehat{T} = \frac{15}{13}a + \frac{6}{13}(b-1).$$

is the one with the smallest variance possible.

Let us now compute this variance. Substituting the values

$$\alpha = \frac{15}{13}, \qquad \beta = \frac{6}{13}, \qquad \gamma = -\frac{6}{13}.$$

into

$$\varepsilon = \widehat{T} - T = x\left(\frac{\alpha}{3} + \frac{4\beta}{3} - 1\right) + y\left(\frac{2\alpha}{3} + \frac{\beta}{2} - 1\right) + \gamma + \alpha e_a + \beta e_b$$

Answers

it follows that

$$\varepsilon = -\frac{6}{13} + \frac{15}{13}e_a + \frac{6}{13}e_b = \frac{15}{13}e_a + \frac{6}{13}(e_b - 1)$$

Owing to the fact that $\mathbb{E}[\varepsilon] = 0$, we finally get:

$$\begin{aligned}\mathbb{E}\left[\varepsilon^2\right] &= \mathbb{E}\left[\left(\frac{15}{13}e_a + \frac{6}{13}(e_b-1)\right)^2\right] \\ &= \frac{225}{169}\underbrace{\mathbb{E}\left[e_a^2\right]}_{\frac{1}{2}} + \frac{180}{169}\underbrace{\mathbb{E}[e_a(e_b-1)]}_{0} + \frac{36}{169}\underbrace{\mathbb{E}\left[(e_b-1)^2\right]}_{\frac{1}{3}} \\ &= \frac{249}{338} \approx 0.74\end{aligned}$$

Specifically:

$$\mathsf{A}:\ a_i = \frac{1}{3}x + \frac{2}{3}y + e_a^{(i)},\ i = 1,2,\ldots,N \quad \mathsf{B}:\ b_j = \frac{4}{3}x + \frac{1}{2}y + e_b^{(j)},\ j = 1,2,\ldots,M$$

where the measurement errors $e_a^{(i)}$ and $e_b^{(j)}$ are **mutually independent random variables** with:

$$\mathbb{E}\left[e_a^{(i)}\right] = 0\,, \quad \mathrm{Var}\left[e_a^{(i)}\right] = \frac{1}{2}; \quad \mathbb{E}\left[e_b^{(j)}\right] = 1\,, \quad \mathrm{Var}\left[e_b^{(j)}\right] = \frac{1}{3}.$$

Consider the estimator

$$\widetilde{T} = \widetilde{\alpha}\sum_{i=1}^{N} a_i + \widetilde{\beta}\sum_{j=1}^{M} b_j + \widetilde{\gamma}$$

where $\widetilde{\alpha}$, $\widetilde{\beta}$, and $\widetilde{\gamma}$ are constant parameters to be determined. Establish the conditions on the parameters $\widetilde{\alpha}$, $\widetilde{\beta}$, $\widetilde{\gamma}$ such that the estimator $\widetilde{T}$ is unbiased.

c) Firstly, we highlight that all the sensors measure the same values of x and y. Hence, the only difference between the various measurements is the realization of stochastic measurement errors. Accordingly, we write:

$$\begin{aligned}\forall i \in \{1,N\}, \quad & a_i = \frac{1}{3}x + \frac{2}{3}y + e_a^{(i)} \\ \forall j \in \{1,M\}, \quad & b_j = \frac{4}{3}x + \frac{1}{2}y + e_b^{(j)}\end{aligned}$$

where $e_a^{(i)}$ and $e_b^{(j)}$ are random variables with the same probabilistic properties of e_a and e_b, respectively.

Using $\widetilde{T}$, the estimation error is given by

$$\begin{aligned}\varepsilon &= \widetilde{T} - T \\ &= \left(\widetilde{\alpha}\sum_{i=1}^{n} a_i + \widetilde{\beta}\sum_{j=1}^{m} b_j + \widetilde{\gamma}\right) - (x+y) \\ &= \widetilde{\alpha}\sum_{i=1}^{N}\left(\frac{1}{3}x + \frac{2}{3}y + e_a^{(i)}\right) + \widetilde{\beta}\sum_{j=1}^{M}\left(\frac{4}{3}x + \frac{1}{2}y + e_b^{(j)}\right) + \widetilde{\gamma} - x - y \\ &= x\left(\frac{N\widetilde{\alpha}}{3} + \frac{4M\widetilde{\beta}}{3} - 1\right) + y\left(\frac{2N\widetilde{\alpha}}{3} + \frac{M\widetilde{\beta}}{2} - 1\right) + \widetilde{\alpha}\sum_{i=1}^{N} e_a^{(i)} + \widetilde{\beta}\sum_{j=1}^{M} e_b^{(j)} + \widetilde{\gamma}\end{aligned}$$

Answers

Therefore, the expected estimation error reads

$$\begin{aligned}\mathbb{E}[\varepsilon] &= \mathbb{E}\left[x\left(\frac{N\widetilde{\alpha}}{3}+\frac{4M\widetilde{\beta}}{3}-1\right)+y\left(\frac{2N\widetilde{\alpha}}{3}+\frac{M\widetilde{\beta}}{2}-1\right)+\widetilde{\alpha}\sum_{i=1}^{N}e_a^{(i)}+\widetilde{\beta}\sum_{j=1}^{m}e_b^{(j)}+\widetilde{\gamma}\right] \\ &= \mathbb{E}[x]\left(\frac{N\widetilde{\alpha}}{3}+\frac{4M\widetilde{\beta}}{3}-1\right)+\mathbb{E}[y]\left(\frac{2N\widetilde{\alpha}}{3}+\frac{M\widetilde{\beta}}{2}-1\right)+\widetilde{\gamma}+\widetilde{\alpha}\sum_{i=1}^{N}\underbrace{\mathbb{E}\left[e_a^{(i)}\right]}_{=0}+\widetilde{\beta}\sum_{j=1}^{M}\underbrace{\mathbb{E}\left[e_b^{(j)}\right]}_{=1} \\ &= \mathbb{E}[x]\left(\frac{N\widetilde{\alpha}}{3}+\frac{4M\widetilde{\beta}}{3}-1\right)+\mathbb{E}[y]\left(\frac{2N\widetilde{\alpha}}{3}+\frac{M\widetilde{\beta}}{2}-1\right)+\widetilde{\gamma}+M\widetilde{\beta}\end{aligned}$$

Hence, the estimator $\widetilde{T}$ is unbiased for every x and y if the following conditions are satisfied:

$$\begin{cases}\dfrac{N\widetilde{\alpha}}{3}+\dfrac{4M\widetilde{\beta}}{3}-1=0 \\ \dfrac{2N\widetilde{\alpha}}{3}+\dfrac{M\widetilde{\beta}}{2}-1=0 \\ \widetilde{\gamma}+M\widetilde{\beta}=0\end{cases}$$

After some trivial algebra, it follows that we can rewrite the conditions for the estimator $\widetilde{T}$ to be unbiased for every x and y as

$$\alpha=\frac{15}{13N}, \qquad \beta=\frac{6}{13M}, \qquad \gamma=-\frac{6}{13}.$$

Hence, the only unbiased estimator $\widetilde{T}$ of the form given in Question 4-c) is given by

$$\bar{z}=\frac{15}{13N}\sum_{i=1}^{N}a_i+\frac{6}{13M}\sum_{j=1}^{M}b_j-\frac{6}{13}=\frac{15}{13N}\sum_{i=1}^{N}a_i+\frac{6}{13}\left(\frac{1}{M}\sum_{j=1}^{M}b_j-1\right).$$

[7 Marks]

Answers